

42.510 Statistical Learning for Data Science

Week 1 - Practice Set (Extended)

Time: 50 minutes

Total: 36 marks

- Answer all questions.
- Show clear reasoning for short-answer questions.
- Use exact fractions where appropriate.
- No calculator is required.

1. Part A: Multiple-choice questions

Question 1

A hospital has records for 50,000 patients. Only 800 records have been manually labeled as “high risk” or “not high risk”; the remaining records contain only measurements such as age, blood pressure, and test results. Which learning setting is most appropriate if the goal is to predict risk labels for future patients?

1. Unsupervised learning
2. Semi-supervised learning
3. Reinforcement learning
4. Generative modeling only

Question 2

Let A and B be events with

$$P(A) = \frac{3}{5}, P(B) = \frac{1}{2}, \text{ and } P(A \cup B) = \frac{4}{5}.$$

What is $P(A \mid B)$?

1. $\frac{1}{5}$
2. $\frac{2}{5}$
3. $\frac{3}{5}$
4. $\frac{4}{5}$

Question 3

A random variable X has distribution $\text{Uniform}(2, 8)$. Which statement is correct?

1. $P(3 < X \leq 6) = \frac{1}{3}$ and $E(X) = 4$
2. $P(3 < X \leq 6) = \frac{1}{2}$ and $E(X) = 5$
3. $P(3 < X \leq 6) = \frac{3}{5}$ and $E(X) = 5$
4. $P(3 < X \leq 6) = \frac{1}{2}$ and $E(X) = 8$

Question 4

Suppose $Y_1, \dots, Y_5$ are i.i.d. Bernoulli(p) observations and the realized sample is

1, 0, 1, 1, 0.

Which expression is the likelihood function for p ?

1. $L(p) = p^2(1 - p)^3$
2. $L(p) = p^3(1 - p)^2$
3. $L(p) = 3p + 2(1 - p)$
4. $L(p) = \binom{5}{3}p^3(1 - p)^2$

Question 5

The sample covariance is $\text{Cov}_{X,Y} = 6$, the sample standard deviations are $s_X = 2$ and $s_Y = 3$.

Let $U = 3X - 2$ and $V = 4 - Y$.

What is the sample correlation between U and V ?

1. -1
2. $-\frac{1}{2}$
3. $\frac{1}{2}$
4. 1

Question 6

A researcher repeatedly takes samples from the same population and constructs a 95% confidence interval for the fixed population mean each time. Which statement is correct?

1. Every computed interval contains 95% of the sampled observations.
2. About 95% of the intervals contain the fixed population mean in the long run.
3. The population mean changes from one repeated sample to another.
4. A realized interval has a 95% probability of containing the fixed mean.

Question 7

A population has known standard deviation $\sigma = 12$. A 95% confidence interval for its mean μ is first constructed using a sample of size $n = 36$. The sample size is then increased to $n = 144$, with the same confidence level and the same population standard deviation.

How does the margin of error change?

1. It is unchanged.
2. It is reduced by a factor of $\frac{1}{2}$.
3. It is reduced by a factor of $\frac{1}{4}$.
4. It is increased by a factor of 2.

2. Part B: Short-answer questions

Question 7

For the observed values

2, 4, 4, 8,

compute the sample mean and sample variance. State clearly which quantity is a parameter and which quantities are statistics.

Question 8

The observations are assumed i.i.d. from a distribution with unknown mean μ and known standard deviation $\sigma = 10$. A sample of size $n = 25$ produces sample mean $|\bar{X}| = 52$.

Using the critical value 1.96, construct a 95% confidence interval for μ . Then give one precise frequentist interpretation of the procedure.

Question 9

Suppose the true parameter is $\theta = 10$. Across four simulated samples, an estimator produces

$T_1 = 8, \quad T_2 = 10, \quad T_3 = 10, \quad T_4 = 12$.

Compute its empirical mean squared error. Also compute its empirical bias and comment briefly on the relationship between bias, variability, and MSE in this example.

Question 10

A company observes five independent customer outcomes:

1, 1, 0, 1, 0,

where 1 denotes a purchase. Assume each outcome follows Bernoulli(p).

1. Write the likelihood and log-likelihood for p .
2. Find the maximum likelihood estimator of p .
3. Explain in one sentence why the order of these five observations does not change the MLE.

Question 11

In the German tank problem, serial numbers are modeled as i.i.d. draws from the discrete uniform distribution on $\{1, 2, \dots, N\}$. A sample of five serial numbers is

58, 61, 26, 99, 46.

Consider the estimators

$$T_1 = X_n,$$

$$T_2 = 2|X| - 1,$$

$$T_3 = X_n + X_1 - 1.$$

Compute all three estimates. Then explain why observing the sample maximum alone does not prove that N equals that maximum.

Question 12

Two regression methods are trained on the same training set.

- Method A has training sum of squared residuals equal to 12 and test sum of squared residuals equal to 9.
 - Method B has training sum of squared residuals equal to 8 and test sum of squared residuals equal to 22.
1. Which method fits the training data better?
 2. Which method appears to predict unseen data better?
 3. Explain why the answers to parts 1 and 2 need not be the same.